

FAKE NEWS FIREWALL

Misinformation as an Epidemic on Social Networks

SEIR MODEL

GRAPH SIGNAL PROCESSING

EPIDEMIC SPREADING

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Graph Signal Processing · Spring 2026

1 Motivation & Project Question

Scale-Free

Network Topology

Hub nodes act as super-spreaders; one bridge node can sever entire downstream clusters.

$R_0 > 1$

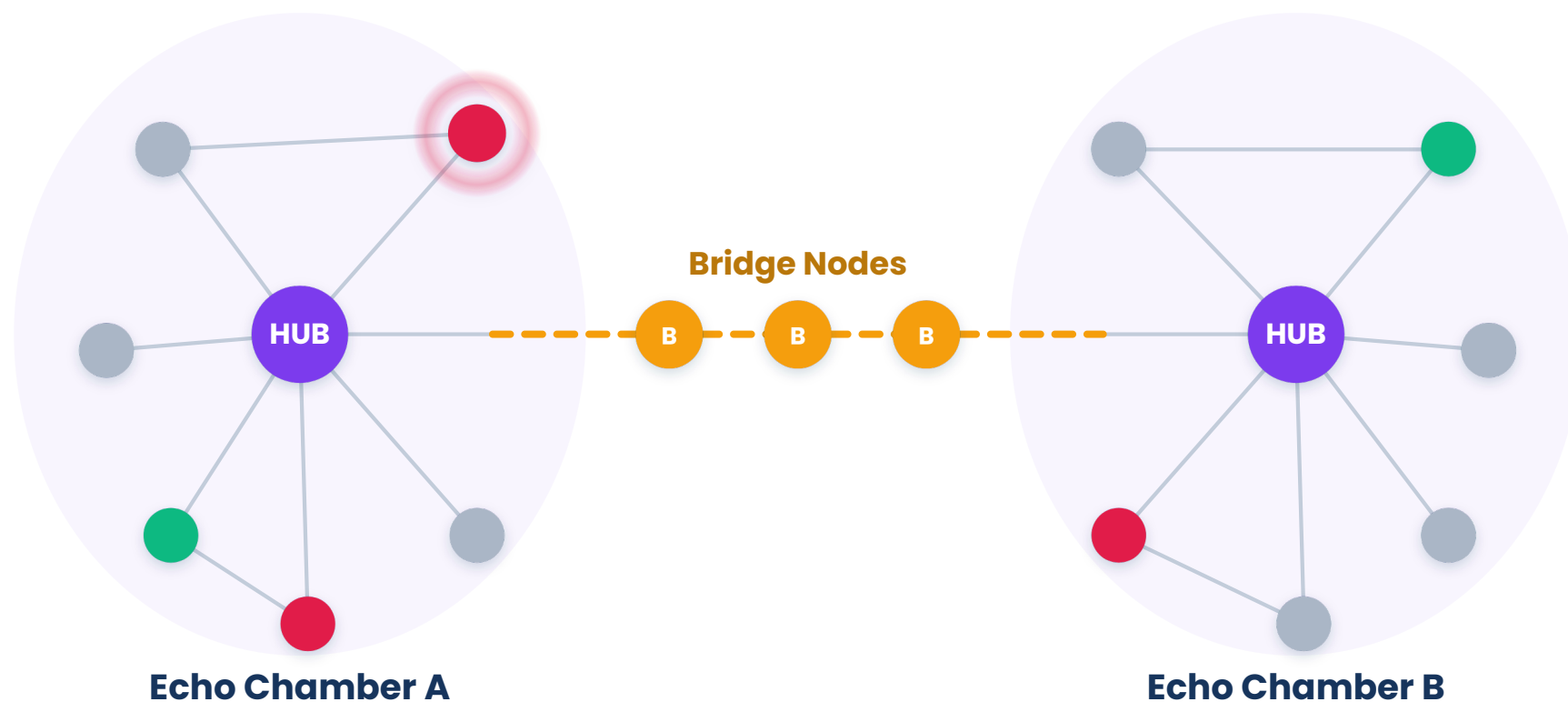
Epidemic Threshold

When the basic reproduction number exceeds 1, misinformation spreads exponentially across clusters.

RESEARCH QUESTION

How does misinformation propagate on scale-free social networks, and which node-level intervention strategy — *degree centrality* (targeting hubs) vs. *betweenness centrality* (targeting bridges) — minimises the final infected fraction under a constrained vaccination budget?

2 Graph Model & Network Structure



- Hub node (high degree)
- Bridge node (high betweenness)
- Infected
- Recovered / Vaccinated
- Susceptible

Barabási-Albert preferential attachment

Louvain community detection

Bridges connect echo chambers

Hub degree: 3× amplified

3 Modified SEIR Epidemic Model



Relapse: $R \rightarrow I$ after tick 10 (mutation event)

β

Transmission Rate

35%

Per tick · hubs 3× amplified

γ

Recovery Rate

20%

Per tick

δ

Relapse Rate

10%

Activates at tick 10

Implementation Stack



NetworkX

BA graph generation, Louvain clustering, betweenness & degree centrality computation.



SEIR Simulation Engine

Discrete-time, tick-based propagation with hub amplification and mutation relapse.



Interactive HTML / JS Demo

Browser-based visualiser: click nodes to vaccinate and watch the epidemic unfold in real time.



Claude AI (Anthropic)

Interactive walkthrough artifact & simulation interface powered via the Anthropic API.

4 Key Insight — Topology Over Content

“It’s not a content problem. It’s a topology problem.”

Where you vaccinate matters more than how much you vaccinate.

STRATEGY A

Degree Centrality

Target the hub — the most connected node in a cluster.

- ✓ Slows spread within one cluster
- ✓ Intuitive — block the influencer
- ✗ Bridge node still carries the story
- ✗ Infection jumps clusters anyway

△ Good, but incomplete

STRATEGY B

Betweenness Centrality

Target the bridge — the cross-cluster bottleneck.

- ✓ Severs the path between echo chambers
- ✓ Entire downstream clusters protected
- ✗ Bridge has few direct connections
- ✗ Requires full-graph analysis to locate

✓ Optimal strategy — block the bridge, not the loudest voice

5 Simulated R_0 Comparison

Final infected fraction after 20 ticks with a 5-token budget. Bar length = infection %.

No intervention $R_0 \approx 3.2$

78%

Degree centrality (hubs) $R_0 \approx 1.9$

51%

Betweenness (bridges) ✓ WIN $R_0 \approx 0.8$

24%

Mixed (bridges + hubs) ✓ BEST $R_0 \approx 0.6$

18%

6 Key Simulation Outcomes

Strategy	Vacc.	Final Inf.	R_0	Outcome
No intervention	0	~78%	3.2	✗ Lose
Random selection	5	~65%	2.7	✗ Lose
Degree centrality (hubs)	5	~51%	1.9	✗ Lose
Betweenness (bridges)	5	~24%	0.8	✓ Win
Mixed (bridges + 1 hub)	5	~18%	0.6	✓ Best

Open Research Questions

RQ 01

Which topologies are most vulnerable? Dense clusters with few bridges are naturally resilient — until the bridge is compromised.

RQ 02

Does betweenness centrality always beat degree centrality? What changes when the graph is denser or sparser?

RQ 03

When recovered nodes can relapse, what is the minimum vaccination budget to prevent a second wave?

RQ 04

Platform design, content labelling and moderation are all network interventions — this simulator makes the trade-offs legible.

“Topology matters more than content.”

Blocking the bridge between echo chambers is more effective than silencing the loudest voice within one.

Gameplay Loop & Implementation

1 Generate Network

Barabási-Albert graph ($N=80, m=2$) with Louvain detection forming 3–4 echo-chamber clusters and bridge nodes.

2 Seed Infection

One hub node starts **Infected**. The story spreads outward along edges every simulation tick.

3 Deploy Fact-Checkers

Place **5 tokens** on chosen nodes, converting them to **Vaccinated** — permanently immune, non-transmitting.

4 Simulate 20 Ticks

SEIR transitions apply each tick. At **tick 10** the mutation event lets recovered nodes relapse (10%).

5 Track R_0 & Final Infection

Win if final infected fraction < 50%. Lose if $\geq 50\%$ of nodes reach Infected.

Limitations & Extensions

△ **Synthetic graph** — real networks have temporal dynamics and directed edges not modelled here.

△ **Homogeneous β** — real sharing rates vary by content type, demographics and platform algorithm.

△ **No counter-narrative agents** — future work could add debunking nodes that spread corrections.

△ **Static community structure** — echo chambers evolve; Louvain is applied once at graph init.

+ **Extension** — weighted edges to model trust relationships between users.

+ **Extension** — multi-layer networks (Twitter + Facebook simultaneously).

Live Demo & References



Try the Simulator

Scan to place fact-checkers on the graph and watch the epidemic evolve in real time.

claude.ai/public/artifacts/defbb637-a3f7-4947-86d7-0a14323dfa9f

🏆 **Goal:** keep infection below 50% using only 5 tokens in 20 ticks. **Hint:** target bridge nodes, not hubs.

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Acknowledgments: Interactive simulator built with Anthropic Claude AI.